

Cellular Biology and Physiological Processes

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Executive Summary

This material covers fundamental biological concepts including cellular structure and function, energy transformation in cells, genetic mechanisms, and physiological regulation. It integrates core principles of biology such as the cell theory, photosynthesis, respiration, and homeostasis.

KEY TAKEAWAYS

The cell is the basic structural and functional unit of all living organisms, and every organism is composed of one or more cells.

Photosynthesis transforms light energy into chemical energy in green plants using the equation $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Cellular respiration produces ATP through three main stages: glycolysis, the Krebs cycle, and the electron transport chain, summarized by the equation $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.

DNA carries genetic instructions and is replicated, transcribed into mRNA, and translated into proteins via ribosomes.

Homeostasis maintains stable internal conditions through feedback mechanisms such as negative and positive feedback.

Enzymes are biological catalysts that speed up chemical reactions and can be denatured by extreme conditions.

Mitosis produces two genetically identical daughter cells, while meiosis leads to genetic diversity in gametes.

Genes and alleles determine traits, with dominant alleles expressed in heterozygotes and recessive alleles only in homozygotes.

Topics

TOPIC 1

Cellular Biology

Defines the cell as the fundamental unit of life and introduces the cell theory and key organelles.

Definition of a Cell

The cell is defined as the basic structural and functional unit of all living organisms.

Every organism is composed of one or more cells.

Cell Theory

The cell theory is a fundamental principle of biology that states that all living things are composed of cells.

New cells are produced from existing cells.

Nucleus

The nucleus is the membrane-bound organelle that contains the genetic material.

Mitochondria

The mitochondria is the organelle responsible for producing ATP through cellular respiration.

TOPIC 2

Photosynthesis

Describes the process by which green plants convert light energy into chemical energy using chloroplasts and chlorophyll.

Definition and Location

Photosynthesis is defined as the process by which green plants transform light energy into chemical energy.

Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll.

Chlorophyll

Chlorophyll is the pigment that absorbs light energy.

Overall Equation

The overall equation for photosynthesis shows that carbon dioxide and water, in the presence of light, produce glucose and release oxygen.

Summary Equation

The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Light-Dependent Reactions

The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.

Calvin Cycle

The Calvin cycle is the light-independent stage that fixes carbon dioxide into glucose.

TOPIC 3

Cellular Respiration

Explains the process of breaking down glucose to produce ATP through three main stages.

Stages of Respiration

Cellular respiration consists of three main stages.

Glycolysis

Glycolysis occurs in the cytoplasm and is the first stage, during which glucose is broken down into two molecules of pyruvate.

Krebs Cycle

The Krebs cycle occurs in the mitochondrial matrix.

Electron Transport Chain

The electron transport chain is the final stage and occurs on the inner mitochondrial membrane.

Fermentation

Fermentation is the anaerobic process that regenerates NAD⁺ when oxygen is unavailable.

Summary Equation

The summary equation of cellular respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.

TOPIC 4

Cell Transport

Describes how substances move across the selectively permeable cell membrane.

Cell Membrane

The cell membrane is the selectively permeable barrier that controls what enters and leaves the cell.

Diffusion

Diffusion is the passive movement of molecules from an area of high concentration to low concentration.

Osmosis

Osmosis is defined as the diffusion of water across a selectively permeable membrane.

Active Transport

Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP.

Endocytosis

Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

TOPIC 5

DNA and Protein Synthesis

Explains how genetic information is stored, replicated, and used to synthesize proteins.

DNA Definition

DNA is defined as the molecule that carries the genetic instructions for the development and functioning of living organisms.

DNA Replication

DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.

Transcription

Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.

Translation

Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence.

Genetic Code

Each three-nucleotide codon on mRNA specifies one amino acid.

The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

TOPIC 6

Cell Division

Describes the process of cell division, including mitosis and the cell cycle.

Mitosis

Mitosis is the process of nuclear division that produces two genetically identical daughter cells.

Cell Cycle

The cell cycle consists of interphase, mitosis, and cytokinesis.

Interphase

Interphase is the stage during which the cell grows, copies its DNA, and prepares for division.

Prophase

Prophase is the first stage of mitosis, during which chromosomes condense.

Anaphase

Anaphase is the stage during which sister chromatids are pulled apart to opposite poles of the cell.

TOPIC 7

Genetics

Explains the principles of heredity, including genes, alleles, and genetic inheritance patterns.

Gene and Allele

A gene is defined as the basic unit of heredity, a segment of DNA that codes for a specific trait.

An allele is an alternative form of a gene at the same locus.

Genotype and Phenotype

The genotype is the genetic makeup of an organism.

The phenotype is its observable characteristics.

Dominant and Recessive Alleles

A dominant allele is expressed when only one copy is present.

A recessive allele is only expressed when two copies are present.

Punnett Squares

Punnett squares are used to predict the probability of offspring inheriting particular traits.

TOPIC 8

Enzymes

Describes the role of enzymes as biological catalysts and factors affecting their function.

Definition and Function

An enzyme is defined as a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.

Active Site

The active site is the region of an enzyme where the substrate binds.

Induced Fit Model

The induced fit model describes how the enzyme changes shape to accommodate the substrate.

Optimal Conditions

Optimal temperature is the temperature at which an enzyme works fastest.

Denaturation

Denaturation is defined as the loss of an enzyme's three-dimensional shape, which destroys its function.

Homeostasis

Explains how organisms maintain stable internal conditions through feedback mechanisms.

Definition

Homeostasis is defined as the maintenance of a stable internal environment despite external changes.

Feedback Mechanisms

Negative feedback is a mechanism that reverses a change, bringing the system back to its set point.

Positive feedback amplifies a change and drives the system further from its set point.

Examples

The normal human body temperature is maintained at approximately 37 degrees Celsius.

Blood glucose concentration is regulated by the hormones insulin and glucagon.

Insulin is secreted when blood glucose is high, promoting glucose uptake.

Human Systems

Describes the structure and functions of major human organ systems.

Circulatory System

The circulatory system transports oxygen, nutrients, and waste products throughout the body.

The heart is the muscular organ that pumps blood through the circulatory system.

Respiratory System

The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.

Nervous System

The nervous system coordinates rapid responses through electrical signals called nerve impulses.

Endocrine System

The endocrine system regulates long-term processes through chemical messengers called hormones.

Terms & Definitions

Term	Definition
Cell	The basic structural and functional unit of all living organisms. Every organism is composed of one or more cells.
Cell theory	A fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells.
Nucleus	The membrane-bound organelle that contains the genetic material.
Mitochondria	The organelle responsible for producing ATP through cellular respiration.
Photosynthesis	The process by which green plants transform light energy into chemical energy.
Chloroplast	The organelle where photosynthesis occurs, containing the green pigment chlorophyll.
Chlorophyll	The pigment that absorbs light energy during photosynthesis.
Light-dependent reactions	Reactions that occur in the thylakoid membranes and produce ATP and NADPH.
Calvin cycle	The light-independent stage of photosynthesis that fixes carbon dioxide into glucose.
C₆H₁₂O₆	$6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
Glycolysis	The first stage of cellular respiration that occurs in the cytoplasm, during which glucose is broken down into two molecules of pyruvate.
Krebs cycle	The stage of cellular respiration that occurs in the mitochondrial matrix.
Electron transport chain	The final stage of cellular respiration that occurs on the inner mitochondrial membrane.
Fermentation	The anaerobic process that regenerates NAD ⁺ when oxygen is unavailable.
Cell membrane	The selectively permeable barrier that controls what enters and leaves the cell.
Diffusion	The passive movement of molecules from an area of high concentration to low concentration.
Osmosis	The diffusion of water across a selectively permeable membrane.
Active transport	The movement of substances against their concentration gradient, requiring energy in the form of ATP.
Endocytosis	The process by which the cell engulfs large particles by wrapping the membrane around them.
DNA	The molecule that carries the genetic instructions for the development and functioning of living organisms.
DNA replication	The process by which a double-stranded DNA molecule is copied to

Term	Definition
	produce two identical DNA molecules.
Transcription	The process by which the DNA sequence of a gene is copied into messenger RNA.
Translation	The process by which the ribosome builds a protein from the mRNA sequence.
mRNA	The molecule that carries the genetic information from DNA to the ribosome for protein synthesis.
Genetic code	The set of rules by which information encoded in genetic material is translated into proteins.
Mitosis	The process of nuclear division that produces two genetically identical daughter cells.
Cell cycle	The cycle consisting of interphase, mitosis, and cytokinesis.
Interphase	The stage during which the cell grows, copies its DNA, and prepares for division.
Prophase	The first stage of mitosis, during which chromosomes condense.
Anaphase	The stage during which sister chromatids are pulled apart to opposite poles of the cell.
Gene	The basic unit of heredity, a segment of DNA that codes for a specific trait.
Allele	An alternative form of a gene at the same locus.
Genotype	The genetic makeup of an organism.
Phenotype	The observable characteristics of an organism.
Dominant allele	An allele that is expressed when only one copy is present.
Recessive allele	An allele that is only expressed when two copies are present.
Punnett square	A tool used to predict the probability of offspring inheriting particular traits.
Enzyme	A protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
Active site	The region of an enzyme where the substrate binds.
Induced fit model	The model describing how the enzyme changes shape to accommodate the substrate.
Optimal temperature	The temperature at which an enzyme works fastest.
Denaturation	The loss of an enzyme's three-dimensional shape, which destroys its function.
Homeostasis	The maintenance of a stable internal environment despite external

Term	Definition
	changes.
Negative feedback	A mechanism that reverses a change, bringing the system back to its set point.
Positive feedback	A mechanism that amplifies a change and drives the system further from its set point.
Circulatory system	The system that transports oxygen, nutrients, and waste products throughout the body.
Heart	The muscular organ that pumps blood through the circulatory system.
Respiratory system	The system responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.
Nervous system	The system that coordinates rapid responses through electrical signals called nerve impulses.
Endocrine system	The system that regulates long-term processes through chemical messengers called hormones.
Insulin	A hormone secreted when blood glucose is high, promoting glucose uptake.
Glucagon	A hormone that regulates blood glucose concentration.
Cellular respiration	$C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$
Pyruvate	A product of glycolysis, formed from the breakdown of glucose.
NAD⁺	A coenzyme that is reduced during cellular respiration and regenerated during fermentation.
NADPH	A product of the light-dependent reactions of photosynthesis, used in the Calvin cycle.
Selectively permeable membrane	A membrane that allows certain substances to pass through while restricting others.
Codon	A three-nucleotide sequence on mRNA that specifies one amino acid.
Chromosomes	Structures composed of DNA and proteins that condense during mitosis.
Sister chromatids	Identical copies of a chromosome produced during DNA replication, separated during anaphase.

Key Facts & Formulas

Formula / Fact	Context
respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.	respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.
$\text{C}_6\text{H}_{12}\text{O}_6$	produce glucose and release oxygen. The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and